# Supplementary Information

Wormhole Formation in Matter-Coupled Vacuum Transitions: Gravitational Collapse, Horizon Constraints, and the Limits of Cosmological Rebirth

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## S1. Equations and scheme

The metric is $ds^2=-f\,du\,dv+r^2d\Omega^2$ with $f=e^{\sigma}$, $G_{\mu\nu}=T_{\mu\nu}$ ($8\pi G=1$) and $T_{\mu\nu}=\partial_\mu\varphi\partial_\nu\varphi-g_{\mu\nu}[(\nabla\varphi)^2/2+U]$. A symbolic computation of the Einstein tensor (derive_equations.py) gives

$$r_{uu}-\sigma_u r_u=-\frac{r}{2}\phi_u^2,\qquad r_{uv}=-\frac{r_u r_v}{r}-\frac{f}{4r}+\frac{Ufr}{4} \tag{S1}$$

$$\phi_{uv}=-\frac{\phi_u r_v+\phi_v r_u}{r}-\frac{f}{4}U'(\phi),\qquad \sigma_{uv}=\frac{f+4r_u r_v}{2r^2}-\phi_u\phi_v \tag{S2}$$

The first of Eqs. (S1) is the constraint along $u$ (that along $v$ is Eq. (4) of the paper) and the others are the evolution equations. The scheme advances rows of constant $u$ on a uniform lattice $u_j=jh$, $v_i=ih$. A cell with corners $W=(j,i-1)$, $A=(j+1,i-1)$, $B=(j,i)$ gives the new point $C=(j+1,i)$ by $X_C=X_A+X_B-X_W+h^2X_{uv}$, with $X_{uv}$ evaluated at the cell centre from the mean values and the central differences, iterated three times (predictor–corrector). Near the centre the evolution of $\sigma$ divides by a small radius and its rounding noise is amplified, so within $K=6$ points of the centre $\sigma$ is integrated from the constraint $\sigma_v=(r_{vv}+r\varphi_v^2/2)/r_v$ (trapezoid), and beyond it from the evolution equation. Each row is solved with the centre value of $\sigma$ as the unknown: elementary flatness requires $\sigma_c=\ln(4r_v^2)$ at the centre, which defines a residual that is solved by the secant method (ten passes at most); an iteration on the plain residual has a gain of modulus greater than one and diverges. The centre value of $\varphi$ follows from $\varphi_t=2\varphi_v$ at the centre by an implicit trapezoid with the one-sided derivative (the explicit iteration has gain $-1.5$).

The data on the initial cone are $r=v/2$, $\varphi(0,v)=A\psi(v)$ and $\sigma(0,v)=\mathrm{int}_0^v\, r\varphi_v^2dv$, the integral of the constraint along $v$ with $f=1$ at the centre. The potential $U=(\kappa/4)(\psi^2-1)^2+s\psi+d$ is tuned by a root finder so that the false minimum has density 3 and the true minimum $-3q$.

## S2. Numerical diagnostics

Misner–Sharp mass of exact solutions: Table 3 of the paper. The diagnostics of each run are the signs of $r_u$ (a forward difference in $u$) and $r_v$ (a central difference in $v$) at every point more than four cells from the centre and eight from the outer boundary; the quadrant map of Fig. 4 is produced by the same classification. The residuals of the constraints are $C_{vv}=r_{vv}-\sigma_v r_v+(r/2)\varphi_v^2$ and $C_{uu}=r_{uu}-\sigma_u r_u+(r/2)\varphi_u^2$, with second differences and central first differences on the lattice.

## S3. Thin-wall check

Table S0 lists the check of the critical mass. For each of nine combinations of $q$ and $\hat{o}$ the finite-tension value $m_c=\max_R m_+(R)$ is computed on a grid of two million points, and the existence of a forbidden window is tested at $m_c(1\mp10^{-3})$.

| $q$ | $\hat{\sigma}$ | $m_c$ | Window at $m_c(1-10^{-3})$ | Window at $m_c(1+10^{-3})$ |
|---|---|---|---|---|
| **0.5** | 0.01 | 1.49999 | yes | no |
| **0.5** | 0.30 | 1.48757 | yes | no |
| **0.5** | 0.90 | 1.39332 | yes | no |
| **1.0** | 0.01 | 1.99998 | yes | no |
| **1.0** | 0.30 | 1.98505 | yes | no |
| **1.0** | 0.90 | 1.86927 | yes | no |
| **2.0** | 0.01 | 2.99998 | yes | no |
| **2.0** | 0.30 | 2.98253 | yes | no |
| **2.0** | 0.90 | 2.84482 | yes | no |

*Table S0. Check of the critical mass (units of $R_{dS}$).*

## S4. Scan of the collapse runs

Supplementary Table S1 lists all runs: field amplitude $A$, wall width $w$, core radius $R_w=V_w/2$, exterior mass $m_{ext}$ (standard deviation across the exterior in parentheses), $R_M=(3m_{ext}/\Delta)^{1/3}$, the radius of the first trapped point at the base resolution, the horizon radius $r_h$ of the measured mass, and the number of anti-trapped points in the core on the last row.

| $A$ | $w$ | $R_w$ | $m_{ext}$ | $R_M$ | $r$ (first trapped) | $r_h$ | Anti-trapped core points |
|---|---|---|---|---|---|---|---|
| **0.05** | 0.05 | 0.40 | 0.166 (0.000) | 0.437 | 0.163 | 0.162 | 0 |
| **0.05** | 0.05 | 0.60 | 0.514 (0.000) | 0.636 | 0.429 | 0.433 | 0 |
| **0.05** | 0.05 | 0.80 | 1.158 (0.000) | 0.834 | 0.743 | 0.745 | 0 |
| **0.05** | 0.05 | 1.00 | 2.188 (0.001) | 1.030 | 1.044 | 1.045 | 0 |
| **0.05** | 0.05 | 1.20 | 3.687 (0.001) | 1.226 | 1.330 | 1.331 | 31 |
| **0.05** | 0.05 | 1.50 | 7.001 (0.001) | 1.518 | 1.738 | 1.739 | 127 |
| **0.05** | 0.05 | 2.00 | 16.033 (0.003) | 2.001 | 2.387 | 2.390 | 287 |
| **0.05** | 0.05 | 3.00 | 51.525 (0.007) | 2.953 | 3.627 | 3.632 | 607 |
| **0.05** | 0.10 | 0.40 | 0.198 (0.000) | 0.462 | — | 0.191 | 0 |
| **0.05** | 0.10 | 0.60 | 0.589 (0.001) | 0.665 | 0.474 | 0.479 | 0 |
| **0.05** | 0.10 | 0.80 | 1.301 (0.001) | 0.867 | 0.791 | 0.796 | 0 |
| **0.05** | 0.10 | 1.00 | 2.426 (0.002) | 1.066 | 1.095 | 1.099 | 0 |
| **0.05** | 0.10 | 1.20 | 4.054 (0.003) | 1.266 | 1.382 | 1.387 | 0 |
| **0.05** | 0.10 | 1.50 | 7.638 (0.006) | 1.563 | 1.795 | 1.801 | 59 |
| **0.05** | 0.10 | 2.00 | 17.393 (0.011) | 2.056 | 2.455 | 2.462 | 159 |
| **0.05** | 0.10 | 3.00 | 55.941 (0.030) | 3.036 | 3.725 | 3.737 | 359 |
| **0.05** | 0.20 | 0.40 | 0.324 (0.001) | 0.545 | — | 0.298 | 0 |
| **0.05** | 0.20 | 0.60 | 0.932 (0.003) | 0.775 | 0.626 | 0.653 | 0 |
| **0.05** | 0.20 | 0.80 | 2.000 (0.006) | 1.000 | 0.968 | 1.000 | 0 |
| **0.05** | 0.20 | 1.00 | 3.656 (0.009) | 1.223 | 1.285 | 1.326 | 0 |
| **0.05** | 0.20 | 1.20 | 6.042 (0.013) | 1.446 | 1.590 | 1.639 | 0 |
| **0.05** | 0.20 | 1.50 | 11.381 (0.021) | 1.785 | 2.039 | 2.101 | 19 |
| **0.05** | 0.20 | 2.00 | 48.383 (4.594) | 2.892 | — | 3.552 | 119 |
| **0.05** | 0.20 | 3.00 | 110.863 (8.554) | 3.813 | — | 4.735 | 319 |
| **0.10** | 0.05 | 0.40 | 0.191 (0.000) | 0.457 | 0.183 | 0.185 | 0 |
| **0.10** | 0.05 | 0.60 | 0.554 (0.000) | 0.652 | 0.456 | 0.458 | 0 |
| **0.10** | 0.05 | 0.80 | 1.194 (0.000) | 0.842 | 0.755 | 0.758 | 0 |
| **0.10** | 0.05 | 1.00 | 2.179 (0.000) | 1.029 | 1.041 | 1.043 | 0 |
| **0.10** | 0.05 | 1.20 | 3.569 (0.001) | 1.213 | 1.310 | 1.312 | 31 |
| **0.10** | 0.05 | 1.50 | 6.535 (0.001) | 1.484 | 1.690 | 1.692 | 127 |
| **0.10** | 0.05 | 2.00 | 14.238 (0.002) | 1.924 | 2.285 | 2.286 | 287 |
| **0.10** | 0.05 | 3.00 | 42.413 (0.007) | 2.768 | 3.389 | 3.392 | 607 |
| **0.10** | 0.10 | 0.40 | 0.203 (0.000) | 0.467 | — | 0.196 | 0 |
| **0.10** | 0.10 | 0.60 | 0.592 (0.001) | 0.666 | 0.480 | 0.481 | 0 |
| **0.10** | 0.10 | 0.80 | 1.285 (0.001) | 0.863 | 0.789 | 0.791 | 0 |
| **0.10** | 0.10 | 1.00 | 2.362 (0.002) | 1.057 | 1.084 | 1.085 | 0 |
| **0.10** | 0.10 | 1.20 | 3.900 (0.003) | 1.249 | 1.362 | 1.364 | 0 |

| $A$ | $w$ | $R_w$ | $m_{ext}$ | $R_M$ | $r$ (first trapped) | $r_h$ | Anti-trapped core points |
|---|---|---|---|---|---|---|---|
| **0.10** | 0.10 | 1.50 | 7.226 (0.006) | 1.534 | 1.760 | 1.761 | 59 |
| **0.10** | 0.10 | 2.00 | 16.052 (0.011) | 2.002 | 2.389 | 2.391 | 159 |
| **0.10** | 0.10 | 3.00 | 49.421 (0.030) | 2.913 | 3.575 | 3.579 | 359 |
| **0.10** | 0.20 | 0.40 | 0.255 (0.000) | 0.503 | — | 0.241 | 0 |
| **0.10** | 0.20 | 0.60 | 0.712 (0.002) | 0.709 | 0.545 | 0.548 | 0 |
| **0.10** | 0.20 | 0.80 | 1.511 (0.004) | 0.911 | 0.864 | 0.865 | 0 |
| **0.10** | 0.20 | 1.00 | 2.739 (0.007) | 1.110 | 1.163 | 1.164 | 0 |
| **0.10** | 0.20 | 1.20 | 4.480 (0.010) | 1.308 | 1.447 | 1.447 | 0 |
| **0.10** | 0.20 | 1.50 | 8.238 (0.017) | 1.603 | 1.854 | 1.855 | 19 |
| **0.10** | 0.20 | 2.00 | 18.222 (0.032) | 2.089 | 2.504 | 2.505 | 119 |
| **0.10** | 0.20 | 3.00 | 56.409 (0.084) | 3.044 | 3.744 | 3.748 | 319 |
| **0.20** | 0.05 | 0.40 | 0.271 (0.000) | 0.514 | 0.248 | 0.254 | 0 |
| **0.20** | 0.05 | 0.60 | 0.671 (0.000) | 0.695 | 0.524 | 0.526 | 0 |
| **0.20** | 0.05 | 0.80 | 1.289 (0.000) | 0.864 | 0.790 | 0.792 | 0 |
| **0.20** | 0.05 | 1.00 | 2.151 (0.000) | 1.025 | 1.033 | 1.037 | 0 |
| **0.20** | 0.05 | 1.20 | 3.286 (0.001) | 1.180 | 1.262 | 1.264 | 31 |
| **0.20** | 0.05 | 1.50 | 5.567 (0.001) | 1.407 | 1.584 | 1.585 | 127 |
| **0.20** | 0.05 | 2.00 | 11.217 (0.003) | 1.777 | 2.089 | 2.090 | 287 |
| **0.20** | 0.05 | 3.00 | 31.893 (0.007) | 2.517 | 3.066 | 3.066 | 607 |
| **0.20** | 0.10 | 0.40 | 0.249 (0.000) | 0.499 | — | 0.236 | 0 |
| **0.20** | 0.10 | 0.60 | 0.661 (0.001) | 0.692 | 0.511 | 0.520 | 0 |
| **0.20** | 0.10 | 0.80 | 1.340 (0.001) | 0.875 | 0.805 | 0.810 | 0 |
| **0.20** | 0.10 | 1.00 | 2.331 (0.002) | 1.052 | 1.074 | 1.078 | 0 |
| **0.20** | 0.10 | 1.20 | 3.674 (0.003) | 1.225 | 1.326 | 1.329 | 0 |
| **0.20** | 0.10 | 1.50 | 6.426 (0.005) | 1.476 | 1.679 | 1.681 | 59 |
| **0.20** | 0.10 | 2.00 | 13.259 (0.011) | 1.879 | 2.226 | 2.226 | 159 |
| **0.20** | 0.10 | 3.00 | 37.225 (0.029) | 2.650 | 3.241 | 3.239 | 359 |
| **0.20** | 0.20 | 0.40 | 0.282 (0.000) | 0.521 | 0.248 | 0.264 | 0 |
| **0.20** | 0.20 | 0.60 | 0.750 (0.001) | 0.721 | 0.553 | 0.567 | 0 |
| **0.20** | 0.20 | 0.80 | 1.535 (0.003) | 0.916 | 0.864 | 0.872 | 0 |
| **0.20** | 0.20 | 1.00 | 2.700 (0.006) | 1.105 | 1.150 | 1.156 | 0 |
| **0.20** | 0.20 | 1.20 | 4.306 (0.009) | 1.291 | 1.419 | 1.423 | 0 |
| **0.20** | 0.20 | 1.50 | 7.655 (0.015) | 1.564 | 1.800 | 1.802 | 19 |
| **0.20** | 0.20 | 2.00 | 16.144 (0.030) | 2.006 | 2.395 | 2.396 | 119 |
| **0.20** | 0.20 | 3.00 | 46.330 (0.078) | 2.851 | 3.501 | 3.499 | 319 |

*Table S1. Collapse runs (q=1, κ=24).*

## S5. Runs without a trapped surface at the base resolution

Table S2 lists the seven runs of Table S1 in which no trapped surface was found at the base resolution: the cause of the stop, the last row reached, the largest $\sigma$ on the last good row and the outcome of refinement by factors of 2 and 4 (the radius of the first trapped point, or none).

| $A$ | $w$ | $R_w$ | $m_{ext}$ | $r_h$ | Stop | $u_{end}$ | $\sigma_{max}$ | $h/2$ | $h/4$ |
|---|---|---|---|---|---|---|---|---|---|
| **0.05** | 0.10 | 0.40 | 0.198 | 0.191 | non-finite values | 0.58 | 2.26 | none | 0.184 |
| **0.05** | 0.20 | 0.40 | 0.324 | 0.298 | non-finite values | 0.58 | 1.54 | none | 0.233 |
| **0.05** | 0.20 | 2.00 | 48.383 | 3.552 | non-finite values | 0.37 | 2.79 | 2.547 | 2.534 |
| **0.05** | 0.20 | 3.00 | 110.863 | 4.735 | non-finite values | 0.29 | 2.70 | 3.834 | 3.821 |
| **0.10** | 0.10 | 0.40 | 0.203 | 0.196 | non-finite values | 0.54 | 1.86 | none | none |
| **0.10** | 0.20 | 0.40 | 0.255 | 0.241 | non-finite values | 0.56 | 1.94 | 0.234 | 0.234 |
| **0.20** | 0.10 | 0.40 | 0.249 | 0.236 | non-finite values | 0.46 | 2.10 | 0.229 | 0.229 |

*Table S2. Runs without a trapped surface at the base resolution (q=1).*

## S6. Exteriors of other vacuum depth

Table S3 lists the runs of Section 5.5 ($A$=0.1): vacuum depth $q$, wall width $w$, core radius $R_w$, exterior mass in units of $m_c=(1+q)R_{dS}$, the horizon radius of the exterior (none above the Nariai mass for $q$<0), the radius of the first trapped point if one was found, the number of anti-trapped core points on the last row and the cause of the stop.

| $q$ | $w$ | $R_w$ | $m_{ext}/m_c$ | $r_h$ | $r$ (first trapped) | Anti-trapped points | Stop |
|---|---|---|---|---|---|---|---|
| **−0.5** | 0.05 | 0.60 | 0.46 | 0.238 | — | 0 | non-finite values |
| **−0.5** | 0.05 | 0.80 | 0.86 | 0.485 | — | 0 | non-finite values |
| **−0.5** | 0.05 | 1.00 | 1.36 | none | — | 0 | non-finite values |
| **−0.5** | 0.05 | 1.20 | 1.95 | none | — | 31 | non-finite values |
| **−0.5** | 0.05 | 1.50 | 2.93 | none | — | 127 | non-finite values |
| **−0.5** | 0.05 | 2.00 | 4.40 | none | — | 287 | non-finite values |
| **−0.5** | 0.05 | 3.00 | 3.48 | none | — | 607 | non-finite values |
| **−0.5** | 0.10 | 0.60 | 0.55 | 0.289 | — | 0 | non-finite values |
| **−0.5** | 0.10 | 0.80 | 1.07 | 0.727 | — | 0 | non-finite values |
| **−0.5** | 0.10 | 1.00 | 1.78 | none | — | 0 | non-finite values |
| **−0.5** | 0.10 | 1.20 | 2.70 | none | — | 0 | non-finite values |
| **−0.5** | 0.10 | 1.50 | 4.46 | none | — | 59 | non-finite values |
| **−0.5** | 0.10 | 2.00 | 8.33 | none | — | 159 | non-finite values |
| **−0.5** | 0.10 | 3.00 | 18.31 | none | — | 359 | non-finite values |
| **0.0** | 0.05 | 0.60 | 0.34 | 0.339 | — | 0 | non-finite values |
| **0.0** | 0.05 | 0.80 | 0.68 | 0.684 | 0.680 | 0 | non-finite values |
| **0.0** | 0.05 | 1.00 | 1.18 | 1.180 | 1.177 | 0 | non-finite values |
| **0.0** | 0.05 | 1.20 | 1.84 | 1.841 | 1.834 | 31 | non-finite values |
| **0.0** | 0.05 | 1.50 | 3.15 | 3.154 | — | 127 | non-finite values |
| **0.0** | 0.05 | 2.00 | 6.21 | 6.213 | — | 287 | non-finite values |
| **0.0** | 0.05 | 3.00 | 15.29 | 15.290 | — | 607 | non-finite values |
| **0.0** | 0.10 | 0.60 | 0.38 | 0.383 | — | 0 | non-finite values |
| **0.0** | 0.10 | 0.80 | 0.79 | 0.786 | — | 0 | non-finite values |
| **0.0** | 0.10 | 1.00 | 1.38 | 1.383 | 1.376 | 0 | non-finite values |
| **0.0** | 0.10 | 1.20 | 2.20 | 2.201 | 2.187 | 0 | non-finite values |
| **0.0** | 0.10 | 1.50 | 3.90 | 3.897 | — | 59 | non-finite values |
| **0.0** | 0.10 | 2.00 | 8.13 | 8.132 | — | 159 | non-finite values |
| **0.0** | 0.10 | 3.00 | 22.58 | 22.580 | — | 359 | non-finite values |
| **2.0** | 0.05 | 0.60 | 0.26 | 0.507 | 0.501 | 0 | non-finite values |
| **2.0** | 0.05 | 0.80 | 0.57 | 0.774 | 0.771 | 0 | non-finite values |
| **2.0** | 0.05 | 1.00 | 1.06 | 1.024 | 1.021 | 0 | non-finite values |
| **2.0** | 0.05 | 1.20 | 1.76 | 1.263 | 1.261 | 31 | non-finite values |
| **2.0** | 0.05 | 1.50 | 3.30 | 1.607 | 1.606 | 127 | non-finite values |
| **2.0** | 0.05 | 2.00 | 7.42 | 2.158 | 2.157 | 287 | non-finite values |
| **2.0** | 0.05 | 3.00 | 23.19 | 3.213 | 3.213 | 607 | non-finite values |
| **2.0** | 0.10 | 0.60 | 0.27 | 0.519 | 0.513 | 0 | non-finite values |
| **2.0** | 0.10 | 0.80 | 0.59 | 0.791 | 0.787 | 0 | non-finite values |
| **2.0** | 0.10 | 1.00 | 1.11 | 1.046 | 1.042 | 0 | non-finite values |
| **2.0** | 0.10 | 1.20 | 1.86 | 1.290 | 1.288 | 0 | non-finite values |
| **2.0** | 0.10 | 1.50 | 3.51 | 1.645 | 1.644 | 59 | non-finite values |
| **2.0** | 0.10 | 2.00 | 7.98 | 2.215 | 2.216 | 159 | non-finite values |
| **2.0** | 0.10 | 3.00 | 25.41 | 3.316 | 3.316 | 359 | non-finite values |

*Table S3. Collapse runs with other exterior vacua (A=0.1).*

## S7. Convergence of the constraints and of the null expansions

Fig. S1 shows the RMS residuals of $C_{vv}$ and $C_{uu}$ against the grid step for the run with a core inside its de Sitter radius ($R_w$=1, region $0.05\le u\le 0.28$, $u+0.4\le v\le 3$) and for the run with a core outside it ($R_w$=2, region $0.05\le u\le 0.15$, $u+0.4\le v\le 4$), and the maximum difference of $r_u$ and $r_v$ at $u$=0.2 (first run) and $u$=0.12 (second) between successive resolutions; Table S4 gives the numbers.

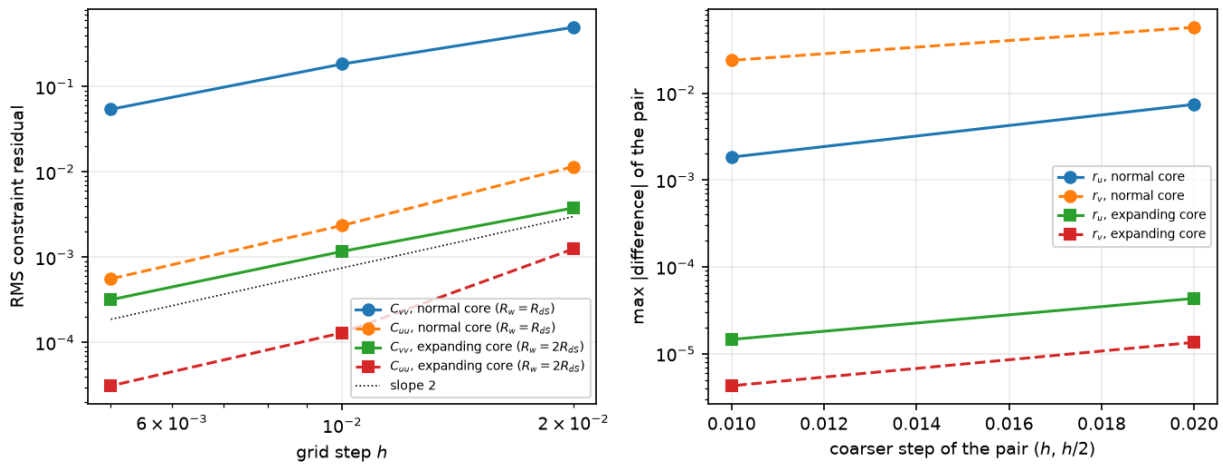



*Fig. S1. (a) RMS residuals of the constraints against the grid step. (b) Maximum difference of the null expansions between successive resolutions.*

| Run | $h$ | Points | RMS $C_{vv}$ | RMS $C_{uu}$ |
|---|---|---|---|---|
| $R_w$=1 | 0.020 | 1,470 | $4.97 \times 10^{-1}$ | $1.16 \times 10^{-2}$ |
| $R_w$=1 | 0.010 | 5,612 | $1.84 \times 10^{-1}$ | $2.36 \times 10^{-3}$ |
| $R_w$=1 | 0.005 | 22,425 | $5.44 \times 10^{-2}$ | $5.59 \times 10^{-4}$ |
| $R_w$=2 | 0.020 | 1,053 | $3.79 \times 10^{-3}$ | $1.25 \times 10^{-3}$ |
| $R_w$=2 | 0.010 | 3,505 | $1.17 \times 10^{-3}$ | $1.30 \times 10^{-4}$ |
| $R_w$=2 | 0.005 | 14,010 | $3.18 \times 10^{-4}$ | $3.12 \times 10^{-5}$ |

*Table S4. RMS constraint residuals of the dynamical solutions.*